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TradingAgents C4 Pipeline Schema

This document maps the end-to-end TradingAgents pipeline as it is wired in this Flint shadow-analysis repo. The repo is an advisory sidecar: it writes evidence artifacts under **output/** and does not write into Flint or place broker orders.

Scope

- Primary entrypoint: [scripts/flint/run_shadow_analysis.py](#)
- Orchestrator: [tradingagents/graph/trading_graph.py](#)
- Graph assembly: [tradingagents/graph/setup.py](#)
- Runtime state: [tradingagents/agents/utils/agent_states.py](#)
- Local artifacts: [output/logs](#), [output/cache](#), [output/memory](#)

C4 Level 1: System Context

```
Parse error on line 1:
C4Contexttitle Trad
^
Expecting 'NEWLINE', 'SPACE', 'GRAPH', got 'ALPHA'
```

C4 Level 2: Containers

```
Parse error on line 1:
C4Containertitle Tr
^
Expecting 'NEWLINE', 'SPACE', 'GRAPH', got 'ALPHA'
```

C4 Level 3: Workflow Components

Parse error on line 1:
C4Componenttitle Tr
^
Expecting 'NEWLINE', 'SPACE', 'GRAPH', got 'ALPHA'

End-to-End Runtime Flow

Parse error on line 2:
...Diagram autonumber participant 0 as
^
Expecting 'SOLID_OPEN_ARROW', 'DOTTED_OPEN_ARROW', 'SOLID_ARROW',
'DOTTED_ARROW', 'SOLID_CROSS', 'DOTTED_CROSS', got 'NL'

Pipeline State Contract

The graph passes one **AgentState** through every node. The important fields are:

Field	Producer	Consumer
company_of_interest, trade_date	Propagator	All agents and tool prompts
past_context	Memory log lookup	Portfolio Manager prompt context
market_report	Market Analyst	Bull/Bear, Trader, Risk analysts
sentiment_report	Social Analyst	Bull/Bear, Trader, Risk analysts
news_report	News Analyst	Bull/Bear, Trader, Risk analysts
fundamentals_report	Fundamentals Analyst	Bull/Bear, Trader, Risk analysts
investment_debate_state	Bull/Bear, Research Manager	Research debate routing and Research Manager
investment_plan	Research Manager	Trader and Portfolio Manager
trader_investment_plan	Trader	Risk analysts and Portfolio Manager
risk_debate_state	Risk analysts, Portfolio Manager	Risk debate routing and Portfolio Manager
final_trade_decision	Portfolio Manager	State logger, memory log, signal parser

Agent And Tool Mapping

Stage	Graph nodes	LLM tier	Tool surface	Output
Market analysis	Market Analyst, tools_market, Msg Clear Market	quick	get_stock_data, get_indicators	market_report
Social analysis	Social Analyst, tools_social, Msg	quick	get_news	sentiment_report

Stage	Graph nodes	LLM tier	Tool surface	Output
	Clear Social			
News analysis	News Analyst, tools_news, Msg Clear News	quick	get_news, get_global_news, get_insider_transactions	news_report
Fundamentals analysis	Fundamentals Analyst, tools_fundamentals, Msg Clear Fundamentals	quick	get_fundamentals, get_balance_sheet, get_cashflow, get_income_statement	fundamentals_report
Investment debate	Bull Researcher, Bear Researcher	quick	none	investment_debate_s
Investment synthesis	Research Manager	deep	none	investment_plan
Trade translation	Trader	quick	none	trader_investment_p
Risk debate	Aggressive Analyst, Conservative Analyst, Neutral Analyst	quick	none	risk_debate_state
Final decision	Portfolio Manager	deep	none	final_trade_decisio
Rating parse	SignalProcessor	deterministic parser	none	Buy, Overweight, Hold Underweight, or Sell

Control Logic

```
Parse error on line 1:
flowchart TD  Start
^
Expecting 'NEWLINE', 'SPACE', 'GRAPH', got 'ALPHA'
```

Artifact Flow

```
Parse error on line 1:
flowchart LR  Env[.
^
Expecting 'NEWLINE', 'SPACE', 'GRAPH', got 'ALPHA'
```

Flint Shadow Boundary

- The wrapper pins `results_dir` to `output/logs`.
- The wrapper pins `data_cache_dir` to `output/cache`.
- The wrapper pins `memory_log_path` to `output/memory/trading_memory.md`.
- The wrapper only prints a JSON summary and writes local artifacts.

- The graph has no broker-order container in this repo-specific model.
- Flint ingestion is a downstream consumer, not a write target from this repo.

C4InterFlow Model

A YAML Architecture-as-Code model for this schema is stored at:

- [Architecture/TradingAgentsFlintShadow.yaml](#)

The prepared render command is:

```
C4InterFlow.Cli draw-diagrams \  
  --aac-input-paths "./Architecture" \  
  --aac-reader-strategy  
"C4InterFlow.Automation.Readers.YamlAaCReaderStrategy,C4InterFlow.Automation" \  
  --interfaces  
"TradingAgentsFlintShadow.SoftwareSystems.*.Containers.*.Components.*.Interfaces.*" \  
  --interfaces "TradingAgentsFlintShadow.SoftwareSystems.*.Containers.*.Interfaces.*" \  
  --types c4 c4-static c4-sequence sequence \  
  --levels-of-details context container component \  
  --formats svg png md \  
  --output-dir "./diagrams/tradingagents-flint-shadow"
```

This command was prepared but not executed here because [C4InterFlow.Cli](#) is not currently installed on PATH.

Source Map

- [scripts/flint/run_shadow_analysis.py](#): wrapper entrypoint, env loading, repo-local path overrides, graph invocation, JSON summary.
- [tradingagents/graph/trading_graph.py](#): graph orchestration, LLM client setup, tool node setup, memory resolution, checkpointing, state logging, memory logging, rating parsing.
- [tradingagents/graph/setup.py](#): LangGraph nodes and edges.
- [tradingagents/graph/conditional_logic.py](#): analyst tool-loop routing, bull/bear debate loop, risk debate loop.
- [tradingagents/graph/propagation.py](#): initial [AgentState](#) construction and graph invocation config.
- [tradingagents/agents/utils/agent_states.py](#): canonical state fields.
- [tradingagents/agents/schemas.py](#): structured [ResearchPlan](#), [TraderProposal](#), and [PortfolioDecision](#) schemas.
- [tradingagents/agents/utils/structured.py](#): structured-output wrapper with free-text fallback.
- [tradingagents/dataflows/interface.py](#): vendor routing and fallback.
- [tradingagents/agents/utils/memory.py](#): append-only decision log and prior-context injection.